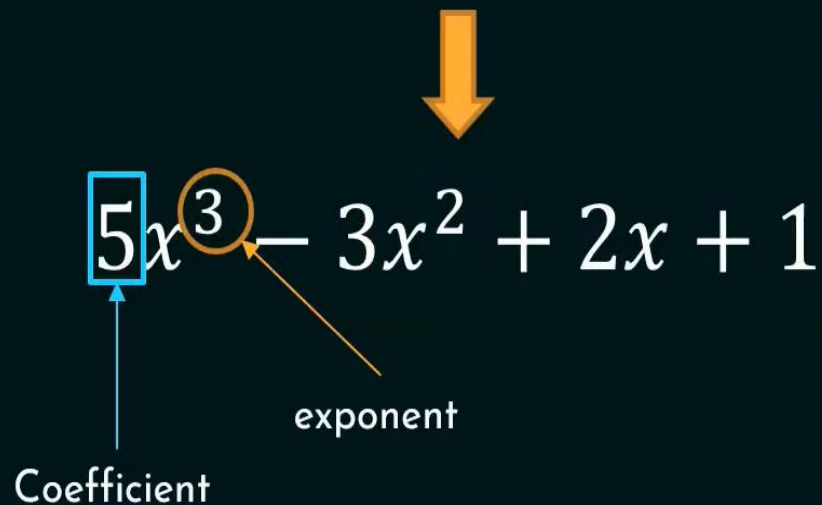


What is polynomial?

Each term of a polynomial has a coefficient and an exponent.



The diagram shows the polynomial expression $5x^3 - 3x^2 + 2x + 1$. A large orange arrow points down from the text above to the first term, $5x^3$. In this term, the coefficient '5' is enclosed in a blue square, and the exponent '3' is enclosed in an orange circle. A blue arrow points from the word 'Coefficient' to the '5'. An orange arrow points from the word 'exponent' to the '3'.

$$5x^3 - 3x^2 + 2x + 1$$

Coefficient

exponent

Each node of the polynomial will represent the term of the expression.

$$5x^3 - 3x^2 + 2x + 1$$

```
struct node {  
    float coefficient;  
    int exponent;  
    struct node* link;  
};
```

5	3	
---	---	--

1000

-3	2	
----	---	--

2000

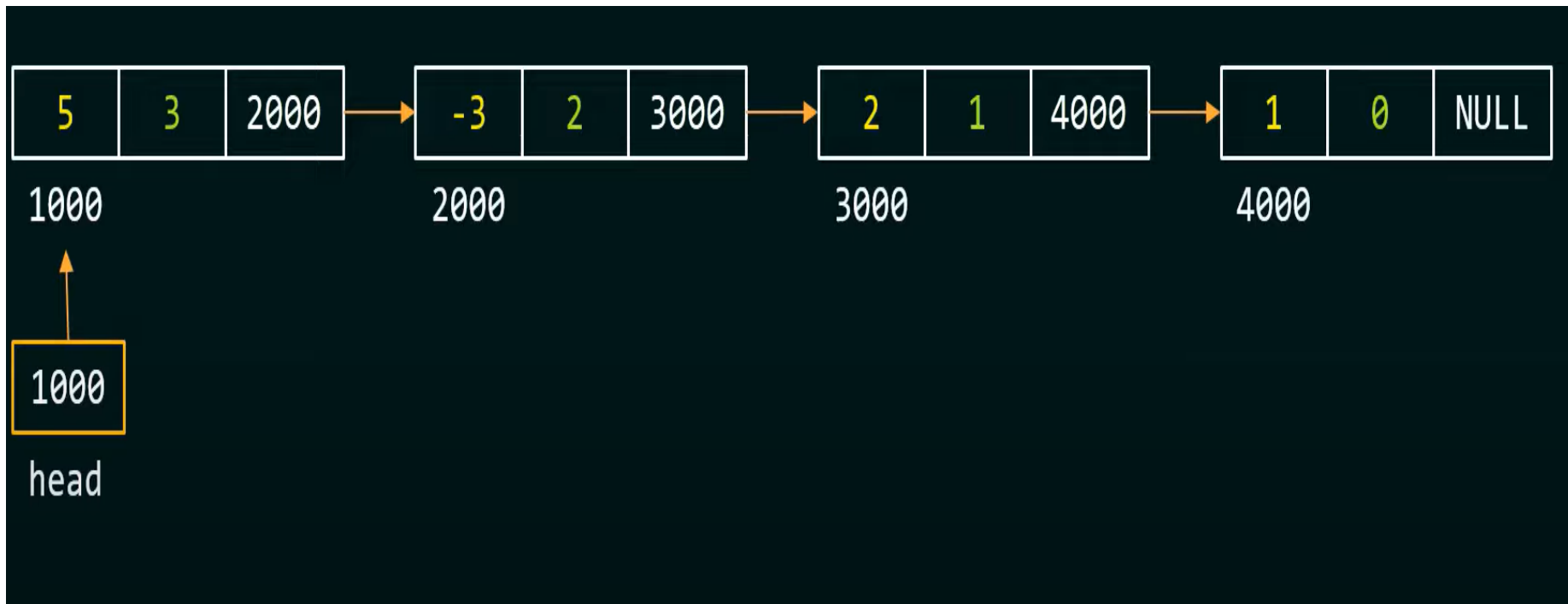
2	1	
---	---	--

3000

1	0	
---	---	--

4000

$$5x^3 - 3x^2 + 2x + 1$$



```
#include<stdio.h>
```

```
#include <stdlib.h>
```

```
struct node
```

```
{
```

```
    float coeff;
```

```
    int expo;
```

```
    struct node* link;
```

```
};
```

```
struct node* create(struct node* head)
{
    int n;
    int i;
    float coeff;
    int expo;
    printf("Enter the number of terms:" );
    scanf("%d", &n);
    for(i=0;i<n;i++)
    {
        printf("Enter the coefficient for term %d:", i+1);
        scanf("%f",&coeff);
        printf("Enter the exponent for for term %d:", i+1);
        scanf("%d",&expo);
        head=insert(head,coeff,expo);
    }
    return head;
}
```

```
struct node* insert(struct node* head, int co, int ex)
{
    struct node* temp;    //Pointer to traverse through nodes
    struct node* newP = malloc(sizeof(struct node)); //New node is created
    newP->coeff=co;
    newP->expo=ex;
    newP->link=NULL;
    if(head == NULL || ex> head->expo)
    {
        newP->link=head;
        head=newP;
    }
    else
    {
        temp=head;
        while(temp->link != NULL && temp->link->expo>=ex)
        {
            temp=temp->link;
        }
        newP->link = temp->link;
        temp->link = newP;
    }
    return head;
}
```

```
void Display(struct node *head)
{
    if(head == NULL)
    {
        printf("No polynomial");
    }
    else
    {
        struct node* temp=head;
        while(temp != NULL)
        {
            printf("%.1fx^%d", temp->coeff, temp->expo);
            temp = temp->link;
            if(temp!=NULL)
                printf("+");
            else
                printf("\n");
        }
    }
}
```

```
int main()  
{  
    struct node* head=NULL;  
    printf("Enter the polynomial\n");  
    head=create(head);  
    Display(head);  
    return 0;  
}
```


Enter the polynomial

Enter the number of terms:3

Enter the coefficient for term 1:5

Enter the exponent for for term 1:3

Enter the coefficient for term 2:3

Enter the exponent for for term 2:2

Enter the coefficient for term 3:1

Enter the exponent for for term 3:0

$(5.0x^3) + (3.0x^2) + (1.0x^0)$

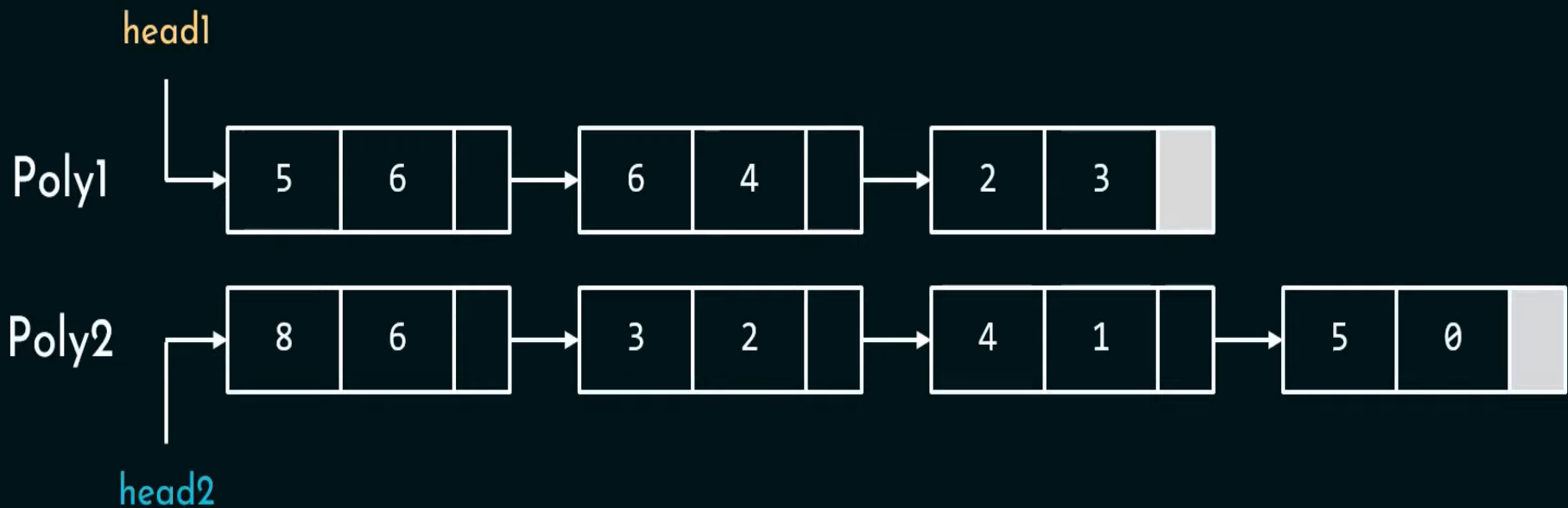
POLYNOMIAL ADDITION

Let say we have the following two polynomials and our job is to add them.

$$5x^6 + 6x^4 + 2x^3$$
$$8x^6 + 3x^2 + 4x + 5$$

Terms are arranged in descending order of their exponents.

Let's represent the above two polynomials using linked lists.



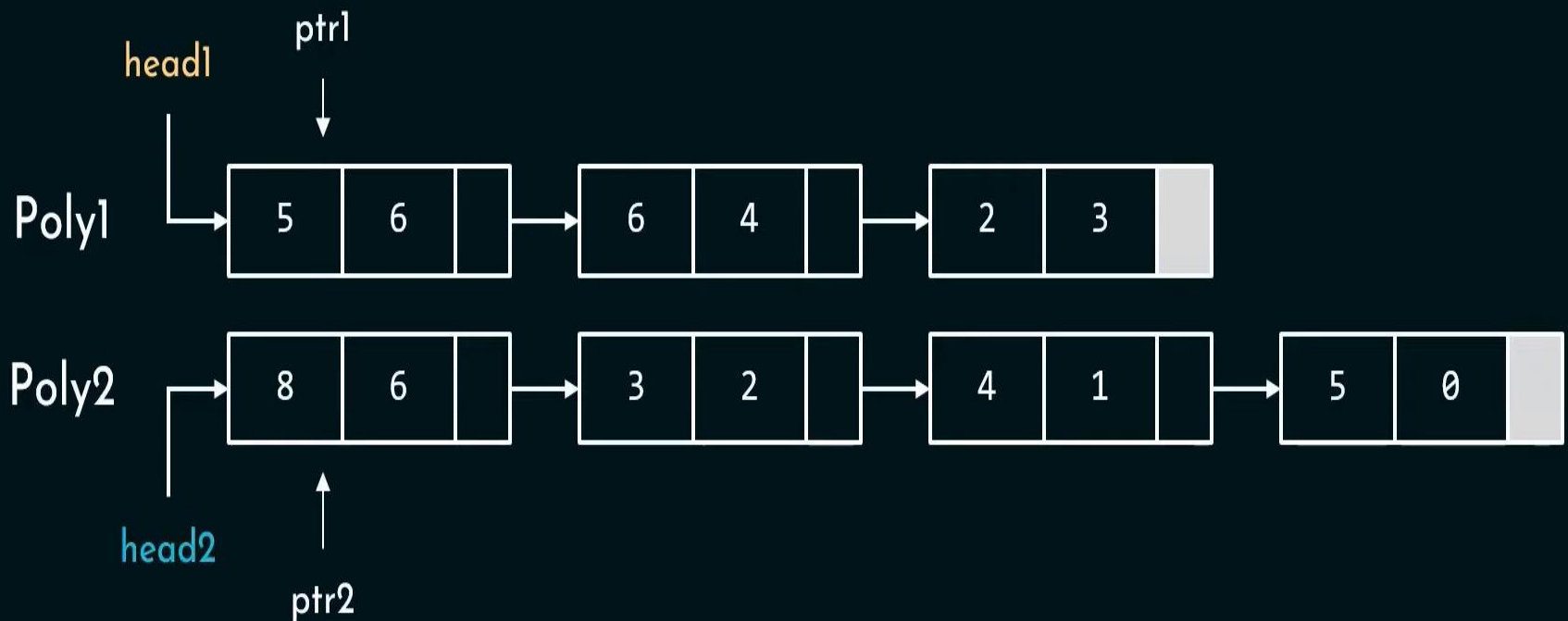
Adding two polynomials means adding their like terms.

The only thing we have to do is to compare their exponents.

$$5x^6 + 6x^4 + 2x^3 \quad (\text{Poly 1})$$

$$8x^6 + 3x^2 + 4x + 5 \quad (\text{Poly 2})$$

$$13x^6 + 6x^4 + 2x^3 + 3x^2 + 4x + 5$$



Algorithm:

Repeat the following until **ptr1** or **ptr2** becomes NULL

if($\text{ptr1} \rightarrow \text{expo} == \text{ptr2} \rightarrow \text{expo}$)

Add the coefficients and insert the newly created node in the resultant linked list and make **ptr1** and **ptr2** point to the next nodes.

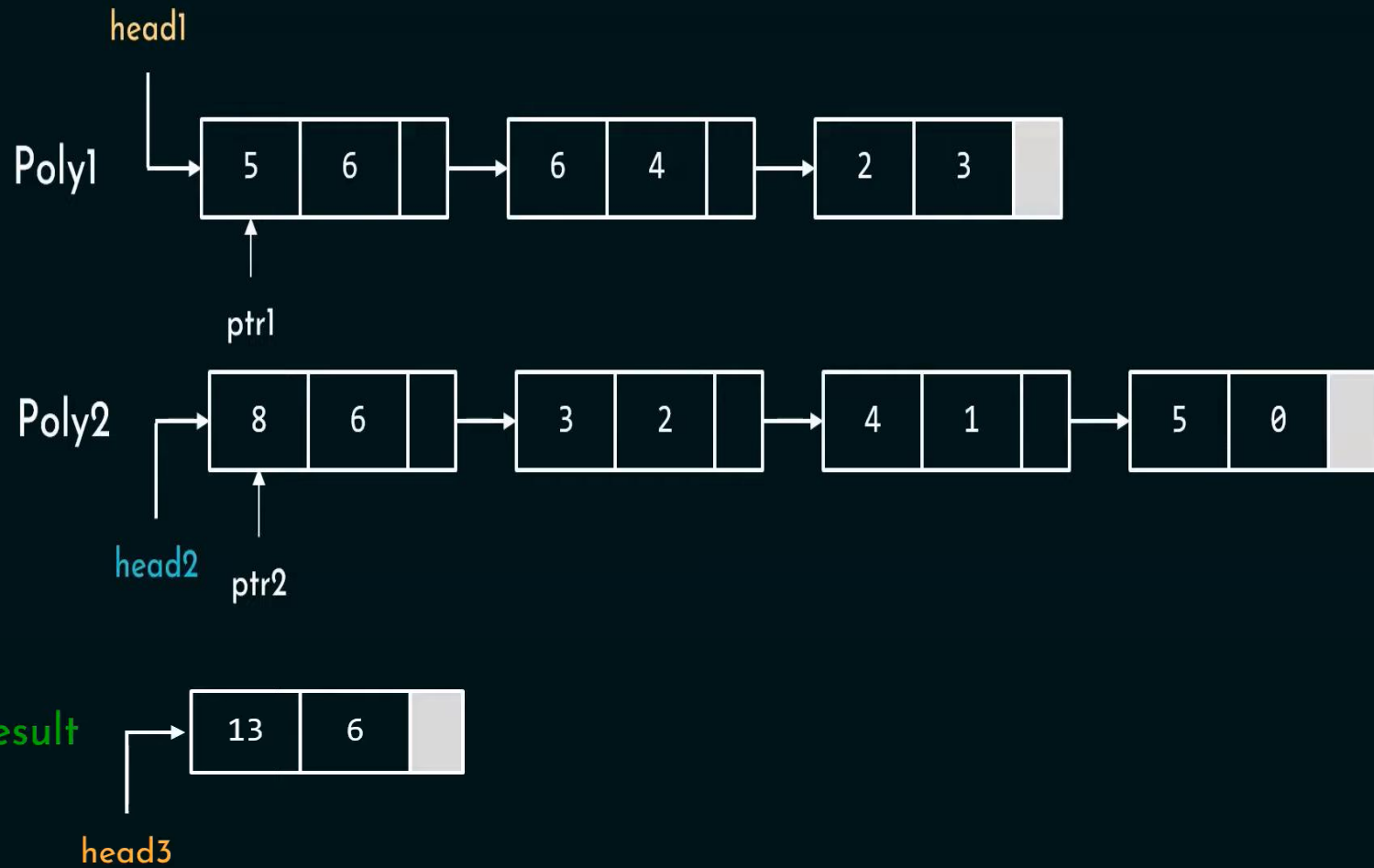
if($\text{ptr1} \rightarrow \text{expo} > \text{ptr2} \rightarrow \text{expo}$)

Insert the node pointed by **ptr1** in the resultant linked list and make **ptr1** point to the next node.

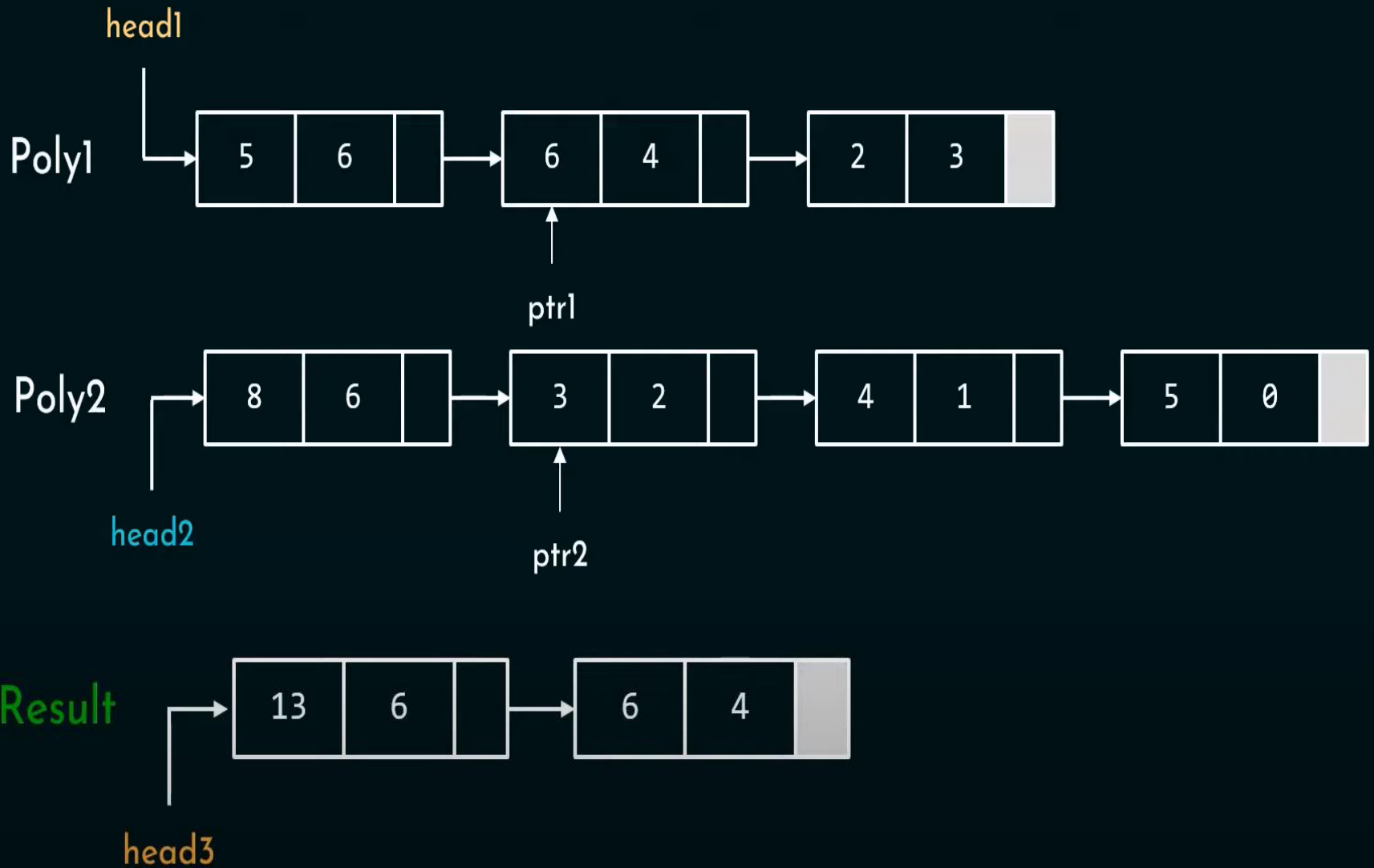
if($\text{ptr1} \rightarrow \text{expo} < \text{ptr2} \rightarrow \text{expo}$)

Insert the node pointed by **ptr2** in the resultant linked list and make **ptr2** point to the next node.

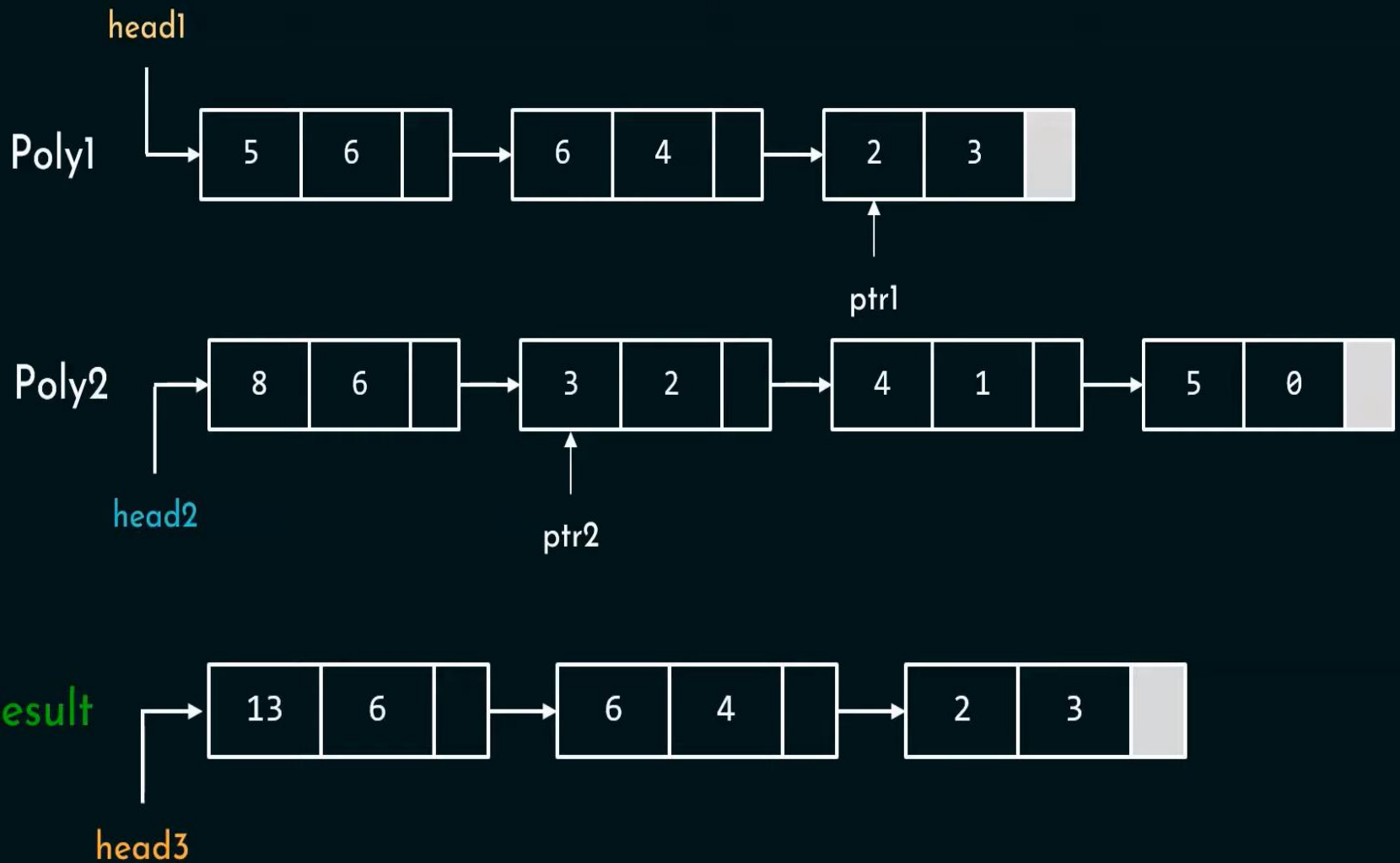
ITERATION 1



ITERATION 2



ITERATION 3



After **iteration 3**, ptr1 becomes NULL.

Now, there is nothing to compare with ptr2->expo.

At this stage, we must add all the remaining nodes of the second linked list at the end of the resultant linked list.

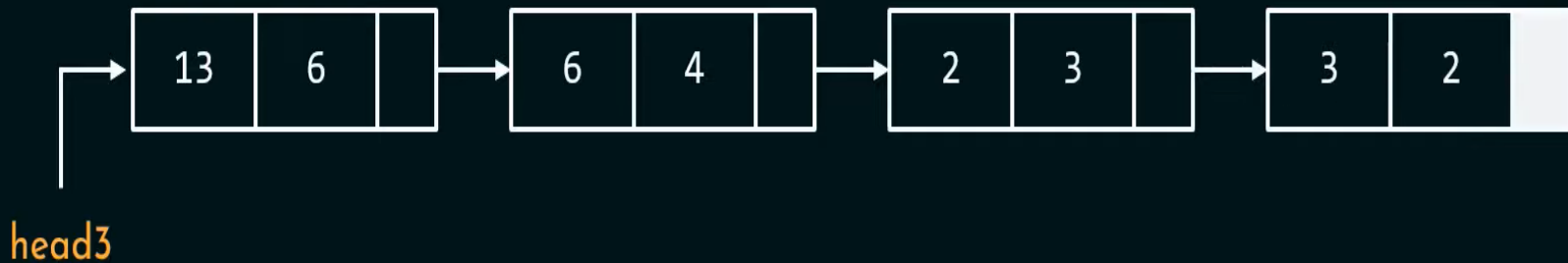
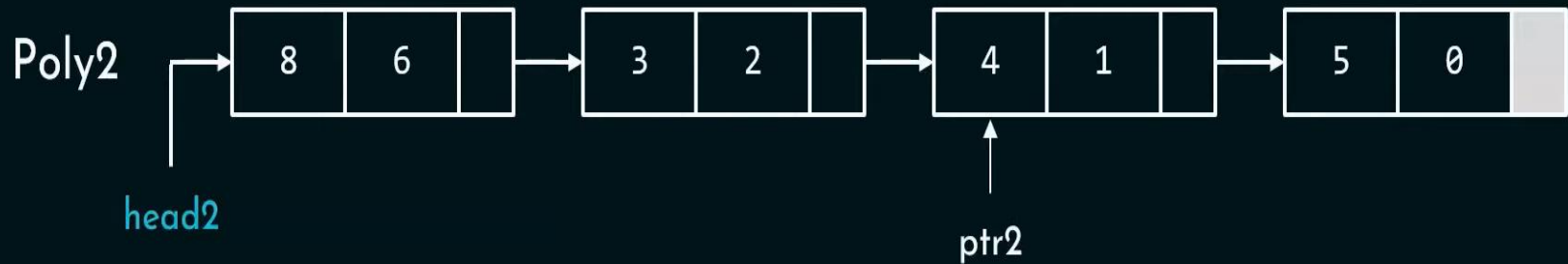
repeat until **ptr2 != NULL**

 Insert the remaining nodes

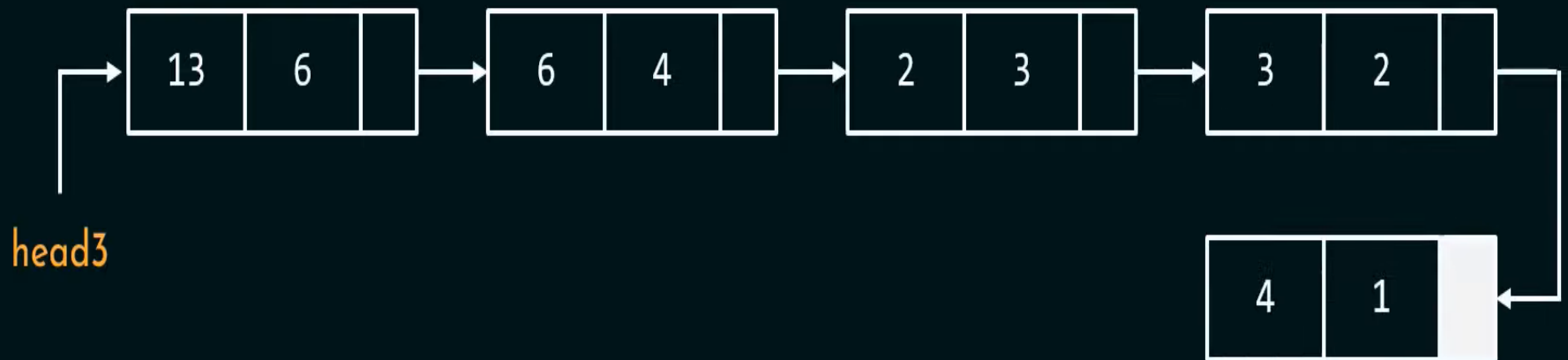
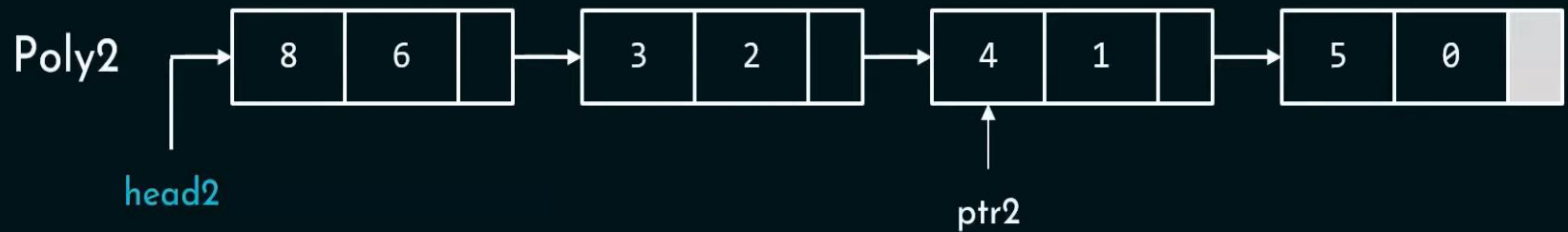
repeat until **ptr1 != NULL**

 Insert the remaining nodes

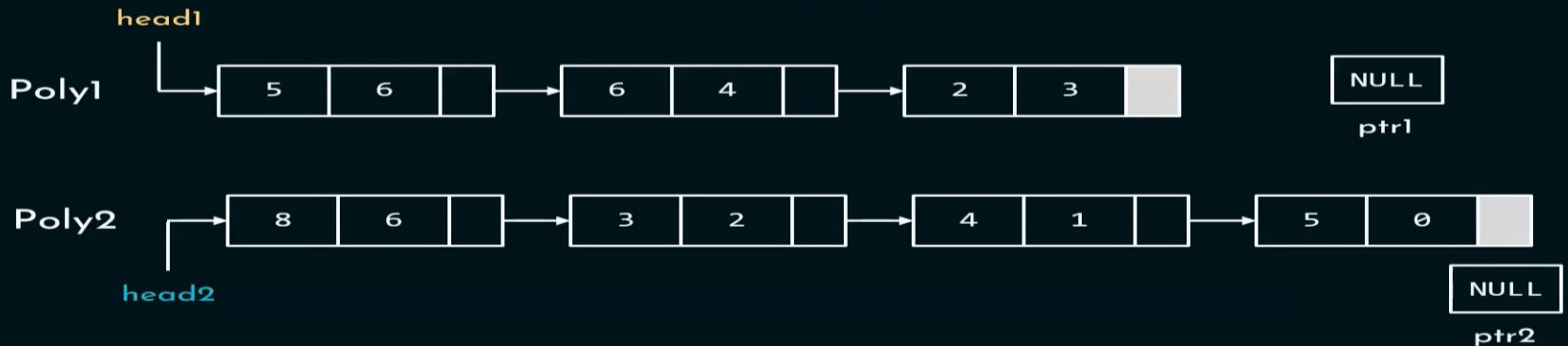
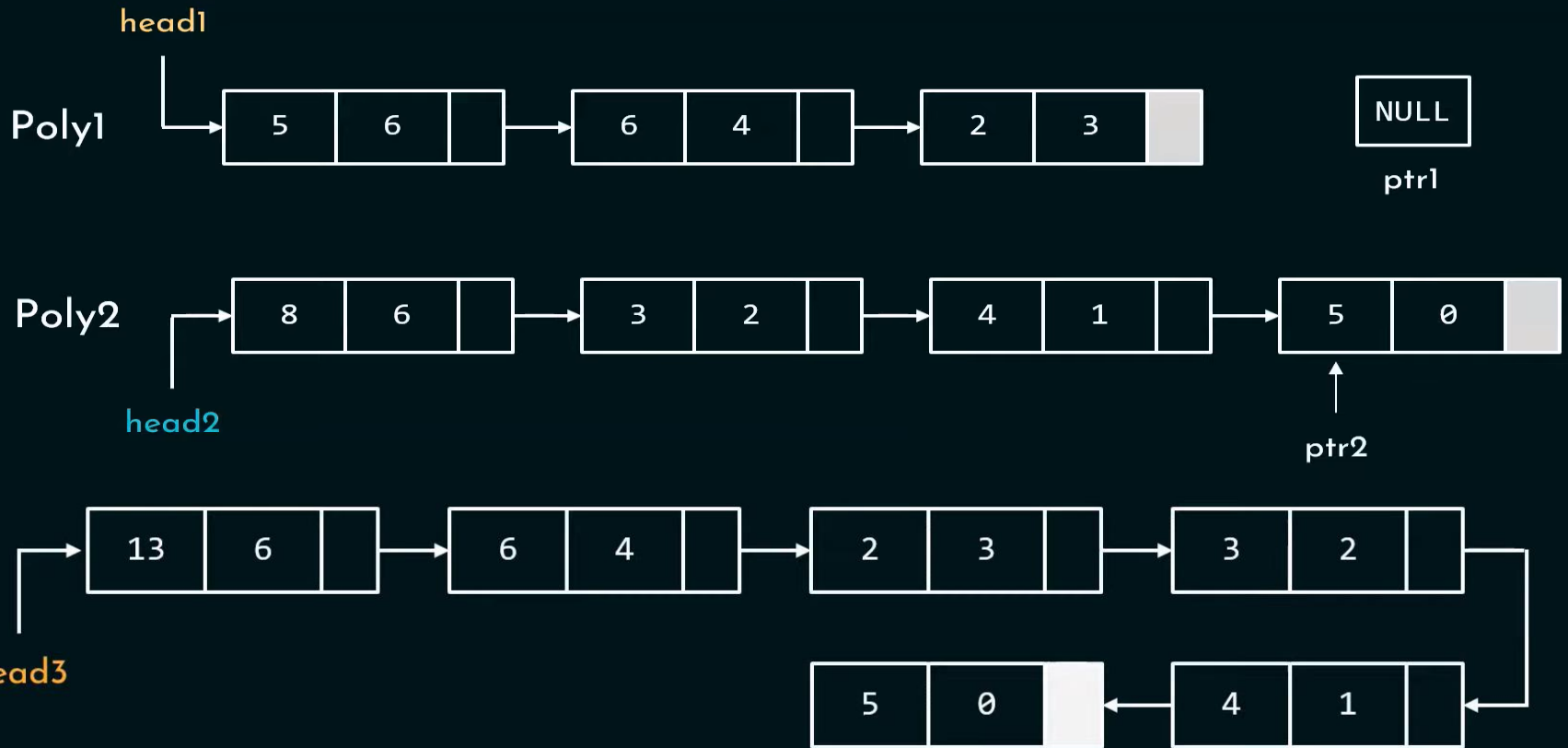
ITERATION 4



ITERATION 5



ITERATION 6



```
int main()  
{  
    struct node* head1=NULL;  
    struct node* head2=NULL;  
    printf("Enter the first polynomial\n");  
    head1=create(head1);  
    printf("Enter the second polynomial\n");  
    head2=create(head2);  
    polyAdd(head1, head2);  
    return 0;  
}
```

```
void polyAdd(struct node* head1, struct node* head2)
{
    struct node* ptr1=head1;
    struct node* ptr2=head2;
    struct node* head3=NULL;
    while(ptr1!=NULL && ptr2!=NULL)
    {
        if(ptr1->expo==ptr2->expo)
        {
            head3=insert(head3,ptr1->coeff+ptr2->coeff,ptr1->expo);
            ptr1=ptr1->link;
            ptr2=ptr2->link;
        }
        else if(ptr1->expo>ptr2->expo)
        {
            head3=insert(head3,ptr1->coeff,ptr1->expo);
            ptr1=ptr1->link;
        }
        else if(ptr1->expo<ptr2->expo)
        {
            head3=insert(head3,ptr2->coeff,ptr2->expo);
            ptr2=ptr2->link;
        }
    }
}
```

```
while(ptr1!=NULL)
{
    head3=insert(head3,ptr1->coeff,ptr1->expo);
    ptr1=ptr1->link;
}
while(ptr2!=NULL)
{
    head3=insert(head3,ptr2->coeff,ptr2->expo);
    ptr2=ptr2->link;
}
printf("Added polynomials:");
Display(head3);
}
```

```
Enter the exponent for for term 1:6
Enter the coefficient for term 2:6
Enter the exponent for for term 2:4
Enter the coefficient for term 3:2
Enter the exponent for for term 3:3
Enter the second polynomial
Enter the number of terms:4
Enter the coefficient for term 1:8
Enter the exponent for for term 1:6
Enter the coefficient for term 2:3
Enter the exponent for for term 2:2
Enter the coefficient for term 3:4
Enter the exponent for for term 3:1
Enter the coefficient for term 4:5
Enter the exponent for for term 4:0
Added polynomials: (13.0x^6)+(6.0x^4)+(2.0x^3)+(3.0x^2)+(4.0x^1)+(5.0x^0)
```


Try on your own!

- Polynomial subtraction
- Polynomial multiplication
- Polynomial division